

Canals Orders API

Manual Test Scenarios — POST /api/v1/orders

v2 — Updated after geocoding mock fix

Supported cities in geocoding mock: Sao Leopoldo, Caxias do Sul, Porto Alegre, Sao Paulo, Rio de Janeiro, New York, San Francisco. Any other city returns 422 GEOCODING_FAILED.

#	Title	Type	Expected Status
1	Successful order — nearest warehouse selected	SUCCESS	201 Created
2	Nearest warehouse lacks product — fallback to Sao Paulo	SUCCESS	201 Created
3	Only the furthest warehouse carries the product	SUCCESS	201 Created
4	Multiple items — warehouse must carry all products	SUCCESS	201 Created
5	Order with complement in shipping address	SUCCESS	201 Created
6	Subtotal and totalAmount calculation correctness	SUCCESS	201 Created
7	Payment declined	ERROR	402 PAYMENT_DECLINED
8	No warehouse has sufficient stock	ERROR	422 WAREHOUSE_NOT_FOUND
9	Invalid customer ID	ERROR	404 CUSTOMER_NOT_FOUND
10	Invalid product ID	ERROR	404 PRODUCT_NOT_FOUND
11	One valid and one invalid product ID	ERROR	404 PRODUCT_NOT_FOUND
12	Empty items array	ERROR	422 EMPTY_ORDER
13	Zod validation — invalid payload fields	ERROR	400 VALIDATION_ERROR
14	Missing required fields — no payment object	ERROR	400 VALIDATION_ERROR
15	Quantity zero — Zod rejects before service	ERROR	400 VALIDATION_ERROR
16	City not recognized by geocoding mock	ERROR	422 GEOCODING_FAILED

#1 — Successful order — nearest warehouse selected

SUCCESS

Expected: 201 Created | warehouse: Porto Alegre Warehouse

Electrical Cable is available at Porto Alegre (100 units). It is the closest warehouse to Sao Leopoldo (~25km).

REQUEST BODY

```
{
  "customerId": "",
  "shippingAddress": {
    "street": "Rua das Flores",
    "number": "100",
    "city": "Sao Leopoldo",
    "state": "RS",
    "country": "Brazil",
    "zipCode": "93010-000"
  },
  "payment": {
```

```
    "cardNumber": "1234567890123456",
    "holderName": "Alice Johnson",
    "expiryDate": "12/27",
    "cvv": "123"
  },
  "items": [
    { "productId": "", "quantity": 1 }
  ]
}
```

EXPECTED RESPONSE

```
{
  "message": "Order created succesfully!",
  "data": {
    "orderId": "",
    "status": "PAID",
    "createdAt": "",
    "warehouse": { "name": "Porto Alegre Warehouse" },
    "totalAmount": 45.99,
    "items": [
      { "productId": "...", "quantity": 1, "unitPrice": 45.99, "subtotal": 45.99 }
    ]
  }
}
```

#2 — Nearest warehouse lacks product — fallback to Sao Paulo

SUCCESS

Expected: 201 Created | warehouse: Sao Paulo Warehouse

Distribution Panel does not exist in Porto Alegre or Caxias. Sao Paulo (~1100km) has 20 units.

REQUEST BODY

```
{
  "customerId": "",
  "shippingAddress": {
    "street": "Rua das Flores",
    "number": "100",
    "city": "Sao Leopoldo",
    "state": "RS",
    "country": "Brazil",
    "zipCode": "93010-000"
  },
  "payment": {
    "cardNumber": "1234567890123456",
    "holderName": "Alice Johnson",
    "expiryDate": "12/27",
    "cvv": "123"
  },
  "items": [
    { "productId": "", "quantity": 1 }
  ]
}
```

EXPECTED RESPONSE

```
{
  "message": "Order created succesfully!",
  "data": {
    "orderId": "",
```

```
"status": "PAID",
"warehouse": { "name": "Sao Paulo Warehouse" },
"totalAmount": 189.9
}
}
```

#3 — Only the furthest warehouse carries the product

SUCCESS

Expected: 201 Created | warehouse: Chicago Warehouse

Step-down Transformer is stocked only at Chicago (~10500km). All others lack it.

REQUEST BODY

```
{
  "customerId": "",
  "shippingAddress": {
    "street": "Av. Brasil",
    "number": "500",
    "city": "Sao Leopoldo",
    "state": "RS",
    "country": "Brazil",
    "zipCode": "93020-000"
  },
  "payment": {
    "cardNumber": "1234567890123456",
    "holderName": "Bob Martinez",
    "expiryDate": "12/27",
    "cvv": "123"
  },
  "items": [
    { "productId": "", "quantity": 1 }
  ]
}
```

EXPECTED RESPONSE

```
{
  "message": "Order created succesfully!",
  "data": {
    "orderId": "",
    "status": "PAID",
    "warehouse": { "name": "Chicago Warehouse" },
    "totalAmount": 320.0
  }
}
```

#4 — Multiple items — warehouse must carry all products

SUCCESS

Expected: 201 Created | warehouse: Sao Paulo Warehouse

Distribution Panel forces Sao Paulo selection, since Porto Alegre and Caxias lack it.

REQUEST BODY

```
{
  "customerId": "",
  "shippingAddress": {
    "street": "Rua das Flores",
    "number": "100",
    "city": "Sao Leopoldo",

```

```
    "state": "RS",
    "country": "Brazil",
    "zipCode": "93010-000"
  },
  "payment": {
    "cardNumber": "1234567890123456",
    "holderName": "Alice Johnson",
    "expiryDate": "12/27",
    "cvv": "123"
  },
  "items": [
    { "productId": "", "quantity": 1 },
    { "productId": "", "quantity": 1 }
  ]
}
```

EXPECTED RESPONSE

```
{
  "message": "Order created succesfully!",
  "data": {
    "orderId": "",
    "status": "PAID",
    "warehouse": { "name": "Sao Paulo Warehouse" },
    "totalAmount": 235.89,
    "items": [
      { "quantity": 1, "unitPrice": 45.99, "subtotal": 45.99 },
      { "quantity": 1, "unitPrice": 189.9, "subtotal": 189.9 }
    ]
  }
}
```

#5 — Order with complement in shipping address

SUCCESS

Expected: 201 Created | complement field preserved in response

Verifies that optional complement field is correctly stored and returned.

REQUEST BODY

```
{
  "customerId": "",
  "shippingAddress": {
    "street": "Honduras Street",
    "number": "300",
    "complement": "Apt 208",
    "city": "Sao Leopoldo",
    "state": "RS",
    "country": "Brazil",
    "zipCode": "93010-000"
  },
  "payment": {
    "cardNumber": "1234567890123456",
    "holderName": "Carol Silva",
    "expiryDate": "12/27",
    "cvv": "123"
  },
  "items": [
    { "productId": "", "quantity": 2 }
  ]
}
```

```
}
```

EXPECTED RESPONSE

```
{
  "message": "Order created succesfully!",
  "data": {
    "shipping": {
      "street": "Honduras Street",
      "number": "300",
      "complement": "Apt 208",
      "city": "Sao Leopoldo",
      "state": "RS",
      "country": "Brazil",
      "zipCode": "93010-000"
    },
    "totalAmount": 24.0
  }
}
```

#6 — Subtotal and totalAmount calculation correctness

SUCCESS

Expected: 201 Created | subtotal and totalAmount match quantity x unitPrice

Cable: 2 x 45.99 = 91.98. Meter: 3 x 75.00 = 225.00. Total: 316.98

REQUEST BODY

```
{
  "customerId": "",
  "shippingAddress": {
    "street": "Rua das Flores",
    "number": "100",
    "city": "Sao Leopoldo",
    "state": "RS",
    "country": "Brazil",
    "zipCode": "93010-000"
  },
  "payment": {
    "cardNumber": "1234567890123456",
    "holderName": "Alice Johnson",
    "expiryDate": "12/27",
    "cvv": "123"
  },
  "items": [
    { "productId": "", "quantity": 2 },
    { "productId": "", "quantity": 3 }
  ]
}
```

EXPECTED RESPONSE

```
{
  "data": {
    "items": [
      { "quantity": 2, "unitPrice": 45.99, "subtotal": 91.98 },
      { "quantity": 3, "unitPrice": 75.0, "subtotal": 225.0 }
    ],
    "totalAmount": 316.98
  }
}
```

#7 — Payment declined

ERROR

Expected: 402 PAYMENT_DECLINED

Card 0000000000000000 always returns declined in the mock.

REQUEST BODY

```
{
  "customerId": "",
  "shippingAddress": {
    "street": "Rua das Flores",
    "number": "100",
    "city": "Sao Leopoldo",
    "state": "RS",
    "country": "Brazil",
    "zipCode": "93010-000"
  },
  "payment": {
    "cardNumber": "0000000000000000",
    "holderName": "Alice Johnson",
    "expiryDate": "12/27",
    "cvv": "123"
  },
  "items": [
    { "productId": "", "quantity": 1 }
  ]
}
```

EXPECTED RESPONSE

```
{
  "error": {
    "code": "PAYMENT_DECLINED",
    "message": "Payment was declined"
  }
}
```

#8 — No warehouse has sufficient stock

ERROR

Expected: 422 WAREHOUSE_NOT_FOUND

Quantity 99999 exceeds every warehouse stock. No candidates returned.

REQUEST BODY

```
{
  "customerId": "",
  "shippingAddress": {
    "street": "Rua das Flores",
    "number": "100",
    "city": "Sao Leopoldo",
    "state": "RS",
    "country": "Brazil",
    "zipCode": "93010-000"
  },
  "payment": {
    "cardNumber": "1234567890123456",
    "holderName": "Alice Johnson",
    "expiryDate": "12/27",
    "cvv": "123"
  }
}
```

```
},
"items": [
  { "productId": "", "quantity": 99999 }
]
}
```

EXPECTED RESPONSE

```
{
  "error": {
    "code": "WAREHOUSE_NOT_FOUND",
    "message": "No warehouse has sufficient stock for all items"
  }
}
```

#9 — Invalid customer ID

ERROR

Expected: 404 CUSTOMER_NOT_FOUND

UUID format is valid but does not exist in the database.

REQUEST BODY

```
{
  "customerId": "000000000-0000-0000-0000-000000000000",
  "shippingAddress": {
    "street": "Rua das Flores",
    "number": "100",
    "city": "Sao Leopoldo",
    "state": "RS",
    "country": "Brazil",
    "zipCode": "93010-000"
  },
  "payment": {
    "cardNumber": "1234567890123456",
    "holderName": "Alice Johnson",
    "expiryDate": "12/27",
    "cvv": "123"
  },
  "items": [
    { "productId": "", "quantity": 1 }
  ]
}
```

EXPECTED RESPONSE

```
{
  "error": {
    "code": "CUSTOMER_NOT_FOUND",
    "message": "Customer not found"
  }
}
```

#10 — Invalid product ID

ERROR

Expected: 404 PRODUCT_NOT_FOUND | missing ID shown in message

The error message must contain the invalid productId for easy debugging.

REQUEST BODY

```
{
  "customerId": "",
  "shippingAddress": {
    "street": "Rua das Flores",
    "number": "100",
    "city": "Sao Leopoldo",
    "state": "RS",
    "country": "Brazil",
    "zipCode": "93010-000"
  },
  "payment": {
    "cardNumber": "1234567890123456",
    "holderName": "Alice Johnson",
    "expiryDate": "12/27",
    "cvv": "123"
  },
  "items": [
    { "productId": "00000000-0000-0000-0000-000000000000", "quantity": 1 }
  ]
}
```

EXPECTED RESPONSE

```
{
  "error": {
    "code": "PRODUCT_NOT_FOUND",
    "message": "The following products were not found:
                00000000-0000-0000-0000-000000000000"
  }
}
```

#11 — One valid and one invalid product ID

ERROR

Expected: 404 PRODUCT_NOT_FOUND | only the invalid ID shown

Partial product match should fail and report only the missing IDs.

REQUEST BODY

```
{
  "customerId": "",
  "shippingAddress": {
    "street": "Rua das Flores",
    "number": "100",
    "city": "Sao Leopoldo",
    "state": "RS",
    "country": "Brazil",
    "zipCode": "93010-000"
  },
  "payment": {
    "cardNumber": "1234567890123456",
    "holderName": "Alice Johnson",
    "expiryDate": "12/27",
    "cvv": "123"
  },
  "items": [
    { "productId": "", "quantity": 1 },
    { "productId": "00000000-0000-0000-0000-000000000000", "quantity": 1 }
  ]
}
```


EXPECTED RESPONSE

```
{
  "error": {
    "code": "PRODUCT_NOT_FOUND",
    "message": "The following products were not found:
                00000000-0000-0000-0000-000000000000"
  }
}
```

#12 — Empty items array

ERROR

Expected: 422 EMPTY_ORDER

An order with no items should be rejected before any warehouse lookup.

REQUEST BODY

```
{
  "customerId": "",
  "shippingAddress": {
    "street": "Rua das Flores",
    "number": "100",
    "city": "Sao Leopoldo",
    "state": "RS",
    "country": "Brazil",
    "zipCode": "93010-000"
  },
  "payment": {
    "cardNumber": "1234567890123456",
    "holderName": "Alice Johnson",
    "expiryDate": "12/27",
    "cvv": "123"
  },
  "items": []
}
```

EXPECTED RESPONSE

```
{
  "error": {
    "code": "EMPTY_ORDER",
    "message": "Order must contain at least one item"
  }
}
```

#13 — Zod validation — invalid payload fields

ERROR

Expected: 400 VALIDATION_ERROR | fieldErrors with all failing fields

customerId not a UUID, cardNumber only 3 digits, expiryDate wrong format, cvv only 2 digits.

REQUEST BODY

```
{
  "customerId": "not-a-uuid",
  "shippingAddress": {
    "street": "",
    "number": "100",
    "city": "SP",
    "state": "RS",
    "country": "Brazil",

```

```

    "zipCode": "93010-000"
  },
  "payment": {
    "cardNumber": "123",
    "holderName": "",
    "expiryDate": "13/27",
    "cvv": "12"
  },
  "items": []
}

```

EXPECTED RESPONSE

```

{
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "Invalid request payload",
    "details": {
      "customerId": ["Invalid uuid"],
      "payment.cardNumber": ["payment.cardNumber must be exactly 16 digits"],
      "payment.expiryDate": ["payment.expiryDate must be in MM/YY format"],
      "payment.cvv": ["payment.cvv must be exactly 3 digits"]
    }
  }
}

```

#14 — Missing required fields — no payment object

ERROR

Expected: 400 **VALIDATION_ERROR**

Payment field is required. Missing it should return a clear validation error.

REQUEST BODY

```

{
  "customerId": "",
  "shippingAddress": {
    "street": "Rua das Flores",
    "number": "100",
    "city": "Sao Leopoldo",
    "state": "RS",
    "country": "Brazil",
    "zipCode": "93010-000"
  },
  "items": [
    { "productId": "", "quantity": 1 }
  ]
}

```

EXPECTED RESPONSE

```

{
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "Invalid request payload",
    "details": {
      "payment": ["Required"]
    }
  }
}

```

#15 — Quantity zero — Zod rejects before service

ERROR

Expected: 400 VALIDATION_ERROR | items[].quantity must be at least 1

quantity: 0 violates .int().min(1) in the Zod schema — rejected before hitting the service.

REQUEST BODY

```
{
  "customerId": "",
  "shippingAddress": {
    "street": "Rua das Flores",
    "number": "100",
    "city": "Sao Leopoldo",
    "state": "RS",
    "country": "Brazil",
    "zipCode": "93010-000"
  },
  "payment": {
    "cardNumber": "1234567890123456",
    "holderName": "Alice Johnson",
    "expiryDate": "12/27",
    "cvv": "123"
  },
  "items": [
    { "productId": "", "quantity": 0 }
  ]
}
```

EXPECTED RESPONSE

```
{
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "Invalid request payload",
    "details": {
      "items": ["items[].quantity must be at least 1"]
    }
  }
}
```

#16 — City not recognized by geocoding mock — NEW

ERROR

Expected: 422 GEOCODING_FAILED

The mock only recognizes: Sao Leopoldo, Caxias do Sul, Porto Alegre, Sao Paulo, Rio de Janeiro, New York, San Francisco. Any other city triggers this error instead of silently returning (0,0).

REQUEST BODY

```
{
  "customerId": "",
  "shippingAddress": {
    "street": "Rua Central",
    "number": "1",
    "city": "Florianopolis",
    "state": "SC",
    "country": "Brazil",
    "zipCode": "88000-000"
  },
  "payment": {
    "cardNumber": "1234567890123456",
  }
}
```

```
    "holderName": "Alice Johnson",
    "expiryDate": "12/27",
    "cvv": "123"
  },
  "items": [
    { "productId": "", "quantity": 1 }
  ]
}
```

EXPECTED RESPONSE

```
{
  "error": {
    "code": "GEOCODING_FAILED",
    "message": "Address could not be geocoded:
                Rua Central, Florianopolis, SC, Brazil"
  }
}
```
